Subject:	Linear Programming 181 & 171
Assessment:	Assignment 3
Due date:	2026-04-25
Total:	50 Marks

Introduction: Video Presentation

In real-world applications, linear programming (LP) problems often involve a large number of decision variables and constraints, making graphical methods insufficient. To address this, more advanced solution techniques are required. The simplex algorithm is a fundamental method used to efficiently solve large-scale LP problems and is widely applied in industries such as logistics, finance, and operations management.

While the simplex algorithm remains highly effective in practice, it is theoretically classified as an exponential-time algorithm in the worst case. More recent developments in optimisation, such as Karmarkar's algorithm, introduced a polynomial-time approach to solving LP problems, offering alternative computational advantages for very large models.

As modern organisations increasingly rely on data-driven decision-making, the ability to formulate, interpret, and solve linear programming models using appropriate methods has become an essential skill. This assignment aims to deepen your understanding of key linear programming algorithms and their practical applications.

Outline

In groups of four students**,** you are required to prepare a short video presentation explaining a selected linear programming / operations research topic in your own words, supported by your own example(s).

Criteria:

- The video presentation should be no more than 10 minutes long.
- The video presentation should include a brief introduction to the selected topic.
- The video presentation should provide a clear and structured explanation of the topic.
- The video presentation should include worked example(s) to demonstrate understanding.

- The video presentation should include a short conclusion summarising key insights.
- The video presentation should be appropriate for an academic audience and suitable for minors.

Topic Selection

1. Each group will select ONE of the following topics for their presentation:
2. Preparing Linear Programming Models for the Two-Phase Simplex Algorithm
3. The Two-Phase Simplex Algorithm: Method and Application
4. Simplex Algorithm Special Cases (e.g., infeasibility, unboundedness, multiple solutions)
5. Linear Programming with Unrestricted-in-Sign Variables
6. Degeneracy and Convergence in the Primal and Two-Phase Simplex Algorithms

Submission Guidelines

- It is your responsibility to ensure that your video is uploaded to the correct platform before the due date.
- If your video file is too large, you may compress the file, ensuring that the quality remains acceptable.
- Late or incorrectly submitted videos may not be accepted, so plan accordingly.

Rubric

Criteria	**Excellent (Full Marks)**	**Good (75% Marks)**	**Satisfactory (50% Marks)**	**Needs Improvement (25% Marks)**	**Marks Allocated**
Introduction *(Clarity, relevance, and engagement in introducing the topic)*	Clearly and concisely introduces the operational research topic. Engaging, well-structured, and sets a strong foundation for	The introduction is clear and relevant but lacks engagement or depth.	The introduction is clear but lacks detail or is not well-structured.	Weak or unclear introduction that does not adequately introduce the topic.	5

	the presentation.				
Content *(Accuracy, depth, clarity, and relevance of explanations and examples)*	Comprehensive, well-researched, and thoroughly explained with strong, relevant examples. Demonstrates excellent understanding of the topic.	Mostly accurate and well-explained, with relevant examples. Some minor gaps in depth or clarity.	Provides basic explanations and examples but lacks depth or has some inaccuracies.	Limited explanations, missing key details, or containing several inaccuracies.	25
Conclusion *(Summary and final thoughts on the topic)*	Summarises key points effectively, provides a strong concluding thought, and reinforces the importance of the topic.	Summarises key points well but lacks a strong closing statement.	This is a basic summary, but there is a weak connection to the overall topic.	Minimal or unclear summary; does not effectively conclude the presentation.	5
Overall Presentation *(Professionalism, clarity of speech, and visual/audio quality)*	Clear, confident delivery with professional presentation. Good pacing, articulation, and visual/audio quality.	Good presentation with minor issues in speech clarity, pacing, or quality.	Some issues with speech clarity, organisation, or visual/audio quality.	Poor delivery, unclear speech, or major visual/audio issues.	5

Engagement & Creativity *(Use of visuals, examples, and overall engagement of the audience)*	Highly engaged with the creative use of visuals, animations, or examples. It keeps the	Engaging with good visuals or examples could improve creativity.	Some effort to engage the audience, but limited use of visuals or examples.	Minimal engagement or creativity. Limited use of supporting materials.	5

	audience interested throughout.				
Adherence to Guidelines *(Length, appropriateness, and submission compliance)*	Fully meets length requirements (within 10 minutes), is appropriate for all viewers, and is correctly submitted.	Slightly over/under the time limit or minor submission issues.	Noticeably too short/long or some issues with appropriateness or submission.	Major issues with length, appropriateness, or submission.	5

Additional Information

- This is a group work project to be completed in groups of 4.
- Belgium Campus has software that can **scan for plagiarism; a** student caught doing this will get a 0 for this assignment.
- **Late assignments or incorrect assignment submissions will not be accepted; missing the deadline is an automatic 0.**